

16.2.2. PID Control Parameters and Clipping Values

PID_DV_CLIP 0x5E

In order to set parameters and clipping values for PID regulation correctly, consider the following details:

Large velocity variations are avoided by limiting v_{PID} value with *PID_DV_CLIP* (register 0x5E). This clipping parameter limits both v_{PID} and *PID_VEL*.

PID_I_CLIP 0x5D (14:0)

The error sum *PID_ISUM* (read out at 0x5B) is generated by the integral term. *PID_ISUM* is limited by setting *PID_I_CLIP* register 0x5D.

- i The maximum value of *PID_I_CLIP* must meet the condition $PID_I_CLIP \leq PID_DV_CLIP / PID_I$.
- i If the error sum *PID_ISUM* is not clipped, it is increased with each time step by $PID_I \cdot PID_E$. This continues as long as the motor does not follow.

PID_D_CLKDIV 0x5D (23:16)

Time scaling for deviation (with respect to error correction periods) is controlled by *PID_D_CLKDIV* register.

- i During error correction, fixed clock frequency $f_{PID_INTEGRAL}$ is valid:

$$f_{PID_INTEGRAL}[Hz] = f_{CLK}[Hz] / 128$$

VEL_ACT_PID

The internal velocity *VEL_ACT_PID* alters actual ramp velocity *VACTUAL*. Two settings are provided:

In case *regulation_modus* = b'11, *VACTUAL* is assigned as pulse generator base value and *VEL_ACT_PID* is calculated by $VEL_ACT_PID = VACTUAL + v_{PID}$.

In case *regulation_modus* = b'10, zero is assigned as pulse generator base value. Now, $VEL_ACT_PID = v_{PID}$ is valid.

PID_TOLERANCE 0x5F

TMC4361 provides the programmable hysteresis *PID_TOLERANCE* for target position stabilization; which avoids oscillations through error correction in case *XACTUAL* is close to the real mechanical position.

The PID controller of TMC4361 is programmable up to approximate 100 kHz update rate (at $f_{CLK} = 16$ MHz). This high speed update rate qualifies PID regulation for motion stabilization.

16.2.3. Enabling PID Regulation

Now that PID control parameters and clipping values are configured, as explained above, PID regulation can be enabled. Two options can be selected.

In order to enable PID control, do as follows:

Action:

OPTION 1: BASE PULSE GENERATOR VELOCITY = 0

- Set *regulation_modus* = b'10 (*ENC_IN_CONF* register 0x07).

OPTION 2: BASE PULSE GENERATOR VELOCITY = VACTUAL

- Set *regulation_modus* = b'11 (*ENC_IN_CONF* register 0x07).

Result:

PID regulation is enabled.

NOTE

- Detailed knowledge of a particular application (including dynamics of mechanics) is necessary for PID controller parameterization.



16.3. Closed-Loop Operation

The closed-loop unit of TMC4361 directly modifies output currents and Step/Dir outputs of the internal step generator; which is dependent on the feedback data. The 2-phase closed-loop control of TMC4361 follows a different approach than field-oriented control (FOC); which is similar to PID control cascades. The ramp generator, which assigns target and velocity, is independent of position control (commutation angle control); which is also independent of current control. Closed-loop operation can only be used in combination with 256 microsteps per fullstep.

16.3.1.

Basic

Closed-Loop Parameters

Closed-loop does not control current values via the internal step generator. The currents values at the SPI output and the Step/Dir outputs are verified using the evaluated difference between internal position *XACTUAL* and external position *ENC_POS*; considering the calibrated offset parameter *CL_OFFSET*.

In order to set parameters and clipping values for closed-loop regulation correctly, consider the following details:

CL_OFFSET 0x59

This register contains the basic offset value between internal and external position during calibration process, which is necessary for closed-loop operation, and offers read-write access. The write access can be used if a defined fixed offset value is preferred, which is verified beforehand.

ENC_POS_DEV 0x52

The continuously updated parameter *ENC_POS_DEV* displays the deviation between *XACTUAL* and *ENC_POS*; considering *CL_OFFSET*.

CL_BETA 0x1C (8:0)

CL_BETA is the maximum commutation angle that is used to compensate an evaluated deviation *ENC_POS_DEV*. In case the deviation reaches *CL_BETA* value, the commutation angle remains stable at this value to follow the overload. Also, *CL_MAX* event is triggered at this point.

CL_TOLERANCE 0x5F (7:0)

This parameter is set to select the tolerance range for position deviation. In case $|ENC_POS_DEV| \leq CL_TOLERANCE$, *CL_FIT_F* lag becomes set. In case a mismatch between internal and external position occurs, *CL_FIT* event is triggered to signify when the mismatch is removed.

CL_DELTA_P 0x5C

CL_DELTA_P is a proportional controller that compensates a detected position deviation between internal and external position. See also Figure 63, page 163. In case $|ENC_POS_DEV| \leq CL_TOLERANCE$, *CL_DELTA_P* is automatically set to 1.0. In case $|ENC_POS_DEV| > CL_TOLERANCE$, the closed-loop unit of TMC4361 multiplies *ENC_POS_DEV* with *CL_DELTA_P* and adds the resulting value to the current *ENC_POS*. Thus, a current commutation angle for higher stiffness position maintenance, which is clipped at *CL_BETA*, is calculated.

- i *CL_DELTA_P* consists of 24 bits. The last 16 bits represent decimal places. The final proportional term is thus calculated by: $p_{PID} = CL_DELTA_P / 65536$.
- i Therefore, the higher p_{PID} the faster the reaction on position deviations.

NOTE:

→ A high p_{PID} term can lead to oscillations that must be avoided.

CL_CYCLE 0x63 (31:16)

In case, one absolute encoder is connected, this value represents the delay time in numbers of clock cycles between two consecutive regulation cycles. It is recommended to adjust this value to the regulation cycle; which is either equal or slower than the encoder request rate. In case incremental ABN encoder is selected, this value is automatically set to fetch the fastest possible regulation rate; which in most cases are five clock cycles.



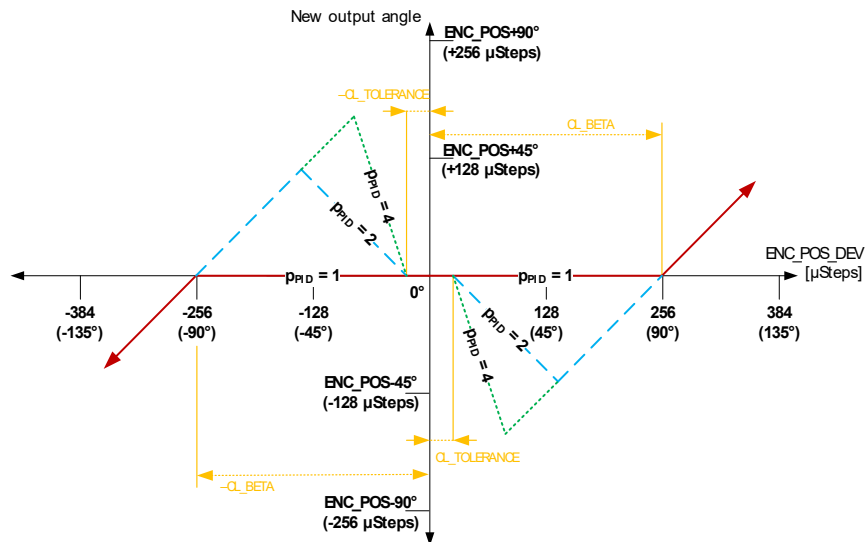


Figure 63: Calculation of the Output Angle with appropriate CL_DELTA_P

16.3.2. Enabling and calibrating Closed-Loop Operation

Now that basic closed-loop control parameters are configured, as explained above, closed-loop regulation can be enabled.

- i The presented calibration process is very basic. Refer to the closed-loop Application Note for detailed calibration process information.

In order to enable and calibrate closed-loop control, do as follows:

PRECONDITION: SET TO BEST POSSIBLE MAXIMUM CURRENT SCALING

PROCEED WITH: OPTION 1: CL OFFSET IS GENERATED DURING CALIBRATION

Action:

- Set $MSTEPS_PER_FS = 0$ ($STEP_CONF$ register 0x0A) [256 microsteps per fullstep].
- Move to any fullstep position ($MSCNT \bmod 128 = 0$).
- Set $regulation_modus = b'01$ (ENC_IN_CONF register 0x07).
- Set $cl_calibration_en = 1$ (ENC_IN_CONF register 0x07).
- Wait for a defined time span (system settle down).
- Set $cl_calibration_en = 0$ (ENC_IN_CONF register 0x07).

Result:

Closed-loop operation is enabled with basic calibration. CL_OFFSET is set to position mismatch during calibration process.

OR PROCEED WITH OPTION 2: CL OFFSET IS USED FOR CALIBRATION

In case CL_OFFSET was saved and no position loss has occurred while closed-loop operation was disabled, it can be used to replace the calibration process.

Action:

- Set $MSTEPS_PER_FS = 0$ ($STEP_CONF$ register 0x0A) [256 microsteps per fullstep].
- Set $regulation_modus = b'01$ (ENC_IN_CONF register 0x07).
- Set CL_OFFSET to any preferred microstep value.

Result:

Closed-loop operation is enabled.



16.3.3. Limiting Closed-Loop Catch-Up Velocity

In order to limit catch-up velocities in case a disturbance of regular motor motion must be compensated, the following parameters can be configured accordingly:

- i Refer to section [16.2.](#) on page [160](#) for more information about PI regulation of the maximum velocity because it uses the same PI regulator like the position PID regulator. The base velocity is the actual ramp velocity *VACTUAL*.

CL_VMAX_CALC_P
0x5A

P parameter of the PI regulator, which controls the maximum velocity.

CL_VMAX_CALC_I
0x5B

I parameter of the PI regulator, which controls the maximum velocity.

PID_DV_CLIP
0x5E

PID_DV_CLIP can be set in order to avoid large velocity variations; and also to limit the maximum velocity deviation above the maximum velocity *VMAX*.

PID_I_CLIP 0x5D

This parameter is used together with *PID_DV_CLIP* in order to limit the velocity for error compensation. The error sum *PID_ISUM* is generated by the integral term. In case this error sum must be limited, set *PID_I_CLIP*.

It is advisable to set the maximum value of *PID_I_CLIP* to:

$$PID_I_CLIP \leq PID_DV_CLIP / PID_I.$$

- i In case the error sum *PID_ISUM* is not clipped, it is increased with each time step by $PID_I \cdot PID_E$. This continues as long as the motor does not follow.

16.3.4. Enabling the Limitation of the Catch-Up Velocity

Now that PI control parameters and clipping values are configured, as explained above, limiting catch-up velocities can be enabled.

In order to enable limitation of closed-loop catch-up velocity, do as follows:

Action:

- Set *cl_vlimit_en* = 1 (*ENC_IN_CONF* register 0x07).

Result:

Closed-loop catch-up velocity is limited according to the configured parameters.

NOTE:

→ A higher motor velocity than specified *VMAX* (for negative velocity: *-VMAX*) is possible if the following conditions are met:

Closed-loop operation is enabled.

Closed-loop catch-up velocity is not enabled, or is enabled with

PID_DV_CLIP > 0; and CL_VMAX_CALC_P and CL_VMAX_CALC_I are higher than 0.

ENC_POS_DEV > CL_TOLERANCE resp. ENC_POS_DEV < CL_TOLERANCE.

AREAS OF SPECIAL CONCERN



In case the internal ramp has stopped, and the position mismatch still needs to be corrected, the base velocity for catch-up velocity limitation is zero.

The mismatch correction ramp is a linear deceleration ramp, independent of the specified ramp profile. This occurs because the catch-up velocity is regulated via PI regulation, as explained above.

Thus, this final ramp for error compensation is a function of both *ENC_POS_DEV* and the PI control parameters.

- → Turn page for information on closed-loop velocity mode.



16.3.5. Enabling Closed- Loop Velocity Mode

Some applications only require maintaining a specified velocity value during closed-loop behavior, regardless of position mismatches. TMC4361 also provides this option.

NOTE:

→ *The closed-loop velocity mode is set independent of the internal ramp operation mode (velocity or positioning mode).*

In order to enable and calibrate closed-loop control, do as follows:

Action:

- Set the catch-up velocity parameters, as explained in detail in section [16.3.3](#), page [164](#).
- Set `cl_vlimit_en` = 1 (`ENC_IN_CONF` register 0x07).
- Set `cl_velocity_mode_en` = 1 (`ENC_IN_CONF` register 0x07).

Result:

Closed-loop operation velocity mode is enabled.

In case position mismatch `|ENC_POS_DEV|` exceeds 768 microsteps, internal position counter `XACTUAL` is set automatically to `ENC_POS ± 768` to limit the position mismatch.

Thus, closed-loop operation maintains the specified velocity value `VMAX`.

- i** A higher motor velocity than specified `VMAX` (for negative velocity: `-VMAX`) is possible if `PID_DV_CLIP` > 0.



16.3.6. Closed-loop Scaling

In order to save energy, current scaling can be adjusted according to actual load during closed-loop operation.

Closed-Loop Scaling Configuration and Enabling

Closed-loop scaling slightly alters the use of the scaling register while remaining consistent in its use of internal scaling and the transmission to the stepper drivers:

1. Closed-loop scaling uses the same scaling register that is also used for open-loop configuration, as explained in chapter 11, page 120. However, the specified values that are used – and thus are also named – differently.
2. Internal scaling of MSLUT current values and transfer of these values to the motor stepper drivers function exactly in the same way as explained in chapter 10, page 88.

In order to configure and enable closed-loop scaling, do as follows:

Action:

- Set proper CL_IMIN ($SCALE_VALUES$ register 0x06).
- Set proper CL_IMAX ($SCALE_VALUES$ register 0x06).
- Set proper CL_START_UP ($SCALE_VALUES$ register 0x06).
- Set $SCALE_VALUES$ (31:24) to 0.
- Set $closed_loop_scale_en = 1$ ($CURRENT_CONF$ register 0x05).

Result:

As soon as closed-loop scaling is enabled, all other open-loop scaling options are automatically disabled. The following scaling situations are possible:

1. In case $|ENC_POS_DEV| \leq CL_START_UP$, current values are scaled with CL_IMIN .
2. In case $|ENC_POS_DEV| > CL_START_UP$ and $|ENC_POS_DEV| \leq CL_BETA$, current values are scaled with a factor that increases linearly from CL_IMIN to CL_IMAX .
3. In case $|ENC_POS_DEV| > CL_BETA$, current values are scaled with CL_IMAX .

The chart below identifies the actual scaling parameter $SCALE_PARAM$, which is dependent on the above described situations:

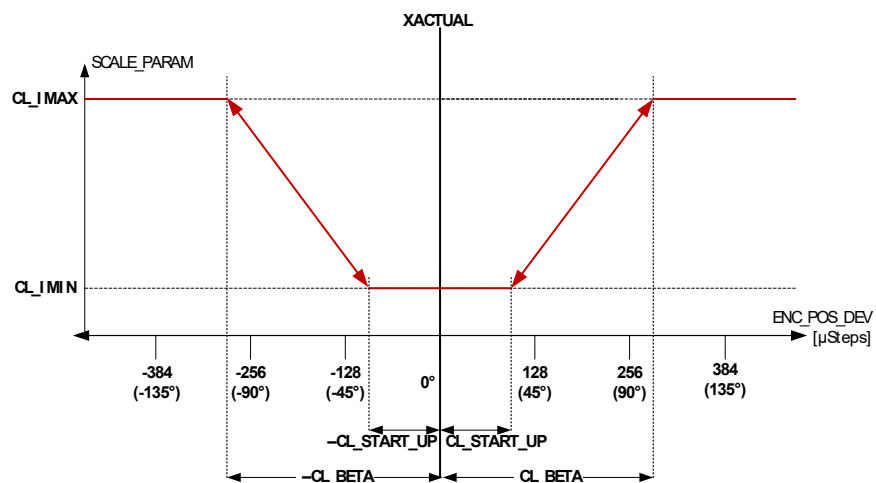


Figure 64: Closed-Loop Current Scaling

•→Turn page for information on closed-loop velocity mode.



16.3.7. Closed-Loop Scaling Transition Process Control

Transition from one scale value to the next active value can be configured as slight conversion. Two different parameters can be set in order to convert to higher or lower closed-loop current scale values, as depicted in the chart below.

In order to configure a smooth transition from a lower motion current scaling value to a higher motion current scaling value, do as follows:

Action:

- Set `CL_UPSCALE_DELAY` register 0x18 according to the delay period after which the actual scale parameter is increased by one step towards the higher current scale value.

Result:

Whenever a higher current scale value is assigned internally, the actual scale parameter is increased by one step per `CL_UPSCALE_DELAY` clock cycles until the assigned scale parameter is reached.

- i If `CL_UPSCALE_DELAY = 0`, the higher current scaling value is immediately assigned whenever the corresponding current scaling phase is activated.

In order to configure a smooth transition from a higher motion current scaling value to a lower motion current scaling value, do as follows:

Action:

- Set `CL_DNSCALE_DELAY` register 0x19 according to the delay period after which the actual scale parameter is decreased by one step towards the lower current scale value.

Result:

Whenever a lower current scale value is assigned internally, the actual scale parameter is decreased by one step per `CL_DNSCALE_DELAY` clock cycles until the assigned scale parameter is reached.

- i If `CL_DNSCALE_DELAY = 0`, the lower current scaling value is immediately assigned whenever the corresponding current scaling phase is activated.

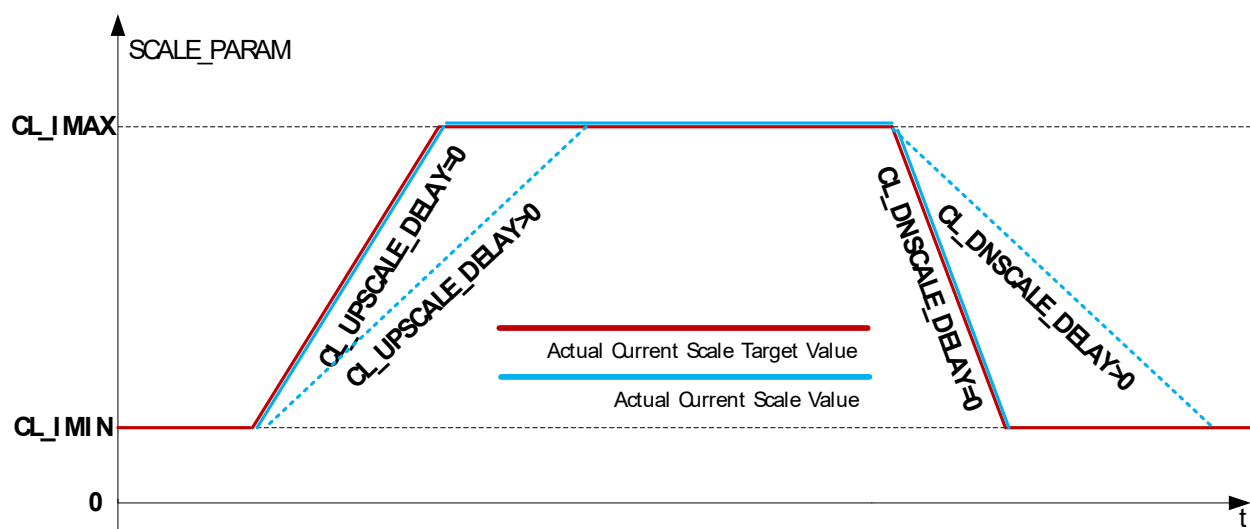


Figure 65: Closed-Loop Current Scaling Timing Behavior



16.3.8. Back-EMF Compensation during Closed-Loop Operation

When higher velocities are reached, a phase shift between current and voltage occurs at the motor coils. Consequently, current control is transformed into voltage control.

This motor- and setup-dependent effect must be compensated because currents are still continuously assigned for motor control. TMC4361 attributes γ -correction to the compensation process, which adds a velocity-dependent angle - *in motion direction* - to the current commutation angle.

Load Angle Calculation

Gamma correction constantly adds one compensation angle, GAMMA, to the actual commutation angle; because the velocity-dependent amount of the influence of Back-EMF, GAMMA is also velocity-dependent. Thus, velocity limits are assigned. These limits are based on REAL motor velocity V_{ENC} (register 0x65). The value of the motor velocity is internally calculated and can be filtered (V_{ENC_MEAN} register 0x66) to smoothen the γ -correction, which is explained in the next section.

In order to configure and enable Back-EMF compensation during closed-loop operation, do as follows:

Action:

- Set proper CL_GAMMA register 0x1C.
- Set proper CL_VMIN_EMF register 0x60.
- Set proper CL_VMAX_EMF register 0x61.
- Set $cl_emf_en = 1$ (ENC_IN_CONF register 0x07).

Result:

Back-EMF compensation during closed-loop operation is enabled. CL_GAMMA represents the maximum value of GAMMA. Per default, CL_GAMMA is set to its maximal possible value of 255, which represents a 90° angle. The following compensation situations are possible:

1. In case $|V_{ENC_MEAN}| \leq CL_VMIN_EMF$, GAMMA is set to 0.
2. In case $|V_{ENC_MEAN}| > CL_VMIN_EMF$ and $|V_{ENC_MEAN}| \leq (CL_VMIN_EMF + CL_VADD_EMF)$, GAMMA is scaled linearly between 0 and its maximum value.
3. In case $|V_{ENC_MEAN}| > (CL_VMIN_EMF + CL_VADD_EMF)$, $GAMMA = CL_GAMMA$.

The chart below identifies the actual parameter GAMMA, which is dependent on the above described situations:

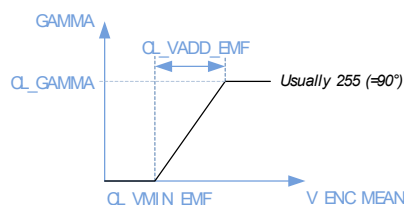


Figure 66: Calculation of the actual Load Angle GAMMA

Areas of Special Concern



If γ -correction is turned on, the maximum possible commutation is $(CL_BETA + CL_GAMMA)$.

This value must not exceed 180° (511 microsteps at 256 microsteps per fullstep) because angles of 180° or more will result in unwanted motion direction changes.



16.3.9. Encoder Velocity Readout Parameters

V_ENC 0x65

In case an encoder is connected, REAL motor velocity can be read out. The actual encoder velocity flickers. This is system-immanent. TMC4361 provides filter options that back-EMF compensation is based on. The following velocity parameters can be read out.

***V_ENC_MEAN
0x66***

Actual filtered encoder velocity in pulses (microsteps) per second [pps].

16.3.10. Encoder Velocity Filter Configuration

***ENC_VMEAN_WAIT
0x63 (7:0)***

In order to set filter parameters correctly, consider the following details:

ENC_VMEAN_WAIT represents the delay period in number of clock cycles between two consecutive *V_ENC* values that are used for the encoder filter velocity calculation. The lower this value, the faster the adaptation process of *V_ENC_MEAN* is. Accordingly: The higher the gradient of *V_ENC_MEAN* is.

In case incremental ABN encoders are connected, *ENC_VMEAN_WAIT* must be set above 32.

In case absolute encoders are connected, *ENC_VMEAN_WAIT* is automatically set to *SER_PTIME*.

***ENC_VMEAN_FILTER
0x63 (11:8)***

This filter exponent is used for filter calculations. The lower this value, the faster the adaptation process of *V_ENC_MEAN* is. Accordingly: The higher the gradient of *V_ENC_MEAN* is. Every *ENC_VMEAN_WAIT* clock cycles, the following calculation applies:

$$V_{ENC_MEAN} = V_{ENC_MEAN} - \frac{V_{ENC_MEAN}}{2^{ENC_VMEAN_FILTER}} + \frac{V_{ENC}}{2^{ENC_VMEAN_FILTER}}$$

***ENC_VMEAN_INT
0x63 (31:16)***

The refresh frequency of high encoder velocity values *V_ENC* is determined by this encoder velocity update period.

In case incremental ABN encoders are connected, the minimum value of *ENC_VMEAN_INT* is automatically set to 256.

In case absolute encoders are connected, *ENC_VMEAN_INT* is automatically adapted to encoder value request rate.

16.3.11. Encoder Velocity equals 0 Event

Because internal calculation of low *V_ENC* values is triggered by AB signal changes and not by the refresh frequency defined by *ENC_VMEAN_INT*, any occurring idle state of the encoder is not recognized.

In order to determine that *V_ENC* = 0, it is possible to limit the number of clock cycles while no AB signal changes occur; which then signifies encoder idle state.

In order to evoke encoder idle state, do as follows:

Action:

- Set proper *ENC_VEL_ZERO* register 0x62.

Result:

In case no AB signal changes occur during *ENC_VEL_ZERO* clock cycles, *ENC_VEL0* event is triggered, which indicates encoder idle state.



TECHNICAL SPECIFICATIONS

17. Complete Register and Switches List

17.1. General Configuration Register GENERAL_CONF 0x00

GENERAL_CONF 0x00 (Default value: 0x00006020)			
R/W	Bit	Val	Remarks
RW	0	<i>use_astart_and_vstart</i> (only valid for S-shaped ramps)	
		0	Sets AACTUAL = AMAX or -AMAX at ramp start and in the case of VSTART ≠ 0.
		1	Sets AACTUAL = ASTART or -ASTART at ramp start and in the case of VSTART ≠ 0.
	1	<i>direct_acc_val_en</i>	
		0	Acceleration values are divided by CLK_FREQ.
		1	Acceleration values are set directly as steps per clock cycle.
	2	<i>direct_bow_val_en</i>	
		0	Bow values are calculated due to division by CLK_FREQ.
		1	Bow values are set directly as steps per clock cycle.
	3	<i>step_inactive_pol</i>	
		0	STPOUT = 1 indicates an active step.
		1	STPOUT = 0 indicates an active step.
	4	<i>toggle_step</i>	
		0	Only STPOUT transitions from inactive to active polarity indicate steps.
		1	Every level change of STPOUT indicates a step.
	5	<i>pol_dir_out</i>	
		0	DIROUT = 0 indicates negative direction.
		1	DIROUT = 1 indicates negative direction.
	7:6	<i>sdin_mode</i>	
		0	Internal step control (internal ramp generator will be used)
		1	External step control via STPIN / DIRIN interface with high active steps at STPIN
		2	External step control via STPIN / DIRIN interface with low active steps at STPIN
		3	External step control via STPIN / DIRIN interface with toggling steps at STPIN
	8	<i>pol_dir_in</i>	
		0	DIRIN = 0 indicates negative direction.
		1	DIRIN = 1 indicates negative direction.
	9	<i>sd_indirect_control</i>	
		0	STPIN/DIRIN input signals will manipulate internal steps at XACTUAL directly.
		1	STPIN/DIRIN input signals will manipulate XTARGET register value, the internal ramp generator is used.
	•→Turn Page.		



GENERAL_CONF 0x00 (Default value: 0x00006020)			
R/W	Bit	Val	Remarks
RW	11:10	<i>serial_enc_in_mode</i>	
		0	An incremental encoder is connected to encoder interface.
		1	An absolute SSI encoder is connected to encoder interface.
		2	Reserved
		3	An absolute SPI encoder is connected to encoder interface.
	12	<i>diff_enc_in_disab</i>	
		0	Differential encoder interface inputs enabled.
		1	Differential encoder interface inputs is disabled (automatically set for SPI encoder).
	14:13	<i>stbby_clk_pin_assignment</i>	
		0	Standby signal becomes forwarded with an active low level at STDBY_CLK output.
		1	Standby signal becomes forwarded with an active high level at STDBY_CLK output.
		2	STDBY_CLK passes ChopSync clock (TMC23x, TMC24x stepper motor drivers only).
		3	Internal clock is forwarded to STDBY_CLK output pin.
	15	<i>intr_pol</i>	
		0	INTR=0 indicates an active interrupt.
		1	INTR=1 indicates an active interrupt.
	16	<i>invert_pol_target_reached</i>	
		0	TARGET_REACHED signal is set to 1 to indicate a target reached event.
		1	TARGET_REACHED signal is set to 0 to indicate a target reached event.
	18:17	Reserved. Set to 0x0.	
	19	<i>fs_en</i>	
		0	Fullstep switchover is disabled.
		1	SPI output forwards fullsteps, if $ VACTUAL > FS_VEL$.
	20	<i>fs_sdout</i>	
		0	No fullstep switchover for Step/Dir output is enabled.
		1	Fullsteps are forwarded via Step/Dir output also if fullstep operation is active.
	22:21	<i>dcstep_mode</i>	
		0	dcStep is disabled.
		1	dcStep signal generation will be selected automatically
		2	dcStep with external STEP_READY signal generation (TMC2130).
		3	dcStep with internal STEP_READY signal generation (TMC26x). i TMC26x config: use const_toff-Chopper (CHM = 1); slow decay only (HSTRRT = 0); TST = 1 and SGT0=SGT1=1 (on_state_xy).
	•→Continued on next page.		



GENERAL_CONF 0x00 (Default value: 0x00006020)			
R/W	Bit	Val	Remarks
RW	23		<i>pwm_out_en</i>
		0	PWM output is disabled. Step/Dir output is enabled at STPOUT/DIROUT.
		1	STPOUT/DIROUT output pins are used as PWM output (PWMA/PWMB).
	24		<i>serial_enc_out_enable</i>
		0	No encoder is connected to SPI output.
		1	SPI output is used as SSI encoder interface to forward absolute SSI encoder data.
	25		<i>serial_enc_out_diff_disable</i>
		0	Differential serial encoder output is enabled.
		1	Differential serial encoder output is disabled.
	26		<i>automatic_direct_sdin_switch_off</i>
		0	VACTUAL=0 & AACTUAL=0 after switching off direct external step control.
		1	VACTUAL = VSTART and AACTUAL = ASTART after switching off direct external step control.
	27		<i>circular_cnt_as_xlatch</i>
		0	The register value of X_LATCH is forwarded at register 0x36.
		1	The register value of REV_CNT (#internal revolutions) is forwarded at register 0x36.
	28		<i>reverse_motor_dir</i>
		0	The direction of the internal SinLUT is regularly used.
		1	The direction of internal SinLUT is reversed
	29		<i>intr_tr_pu_pd_en</i>
		0	INTR and TARGET_REACHED are outputs with strongly driven output values..
		1	INTR and TARGET_REACHED are used as outputs with gated pull-up and/or pull-down functionality.
	30		<i>intr_as_wired_and</i>
		0	INTR output function is used as Wired-Or in case of <i>intr_tr_pu_pd_en</i> = 1.
		1	INTR output function is used as Wired-And in case of <i>intr_tr_pu_pd_en</i> = 1.
	31		<i>tr_as_wired_and</i>
		0	TARGET_REACHED output function is used as Wired-Or in the case of <i>intr_tr_pu_pd_en</i> = 1.
		1	TARGET_REACHED output function is used as Wired-And in the case of <i>intr_tr_pu_pd_en</i> = 1.

Table 60: General Configuration 0x00



17.2. Reference Switch Configuration Register REFERENCE_CONF 0x01

REFERENCE_CONF 0x01 (Default value: 0x00000000)			
R/W	Bit	Val	Remarks
RW	0	stop_left_en	
		0	STOPL signal processing disabled.
		1	STOPL signal processing enabled.
	1	stop_right_en	
		0	STOPR signal processing disabled.
		1	STOPR signal processing enabled.
	2	pol_stop_left	
		0	STOPL input signal is low active.
		1	STOPL input signal is high active.
	3	pol_stop_right	
		0	STOPR input signal is low active.
		1	STOPR input signal is high active.
	4	invert_stop_direction	
		0	STOPL/STOPR stops motor in negative/positive direction.
		1	STOPL/STOPR stops motor in positive/negative direction.
	5	soft_stop_en	
		0	Hard stop enabled. VACTUAL is immediately set to 0 on any external stop event.
		1	Soft stop enabled. A linear velocity ramp is used for decreasing VACTUAL to v = 0.
	6	virtual_left_limit_en	
		0	Position limit VIRT_STOP_LEFT disabled.
		1	Position limit VIRT_STOP_LEFT enabled.
	7	virtual_right_limit_en	
		0	Position limit VIRT_STOP_RIGHT disabled.
		1	Position limit VIRT_STOP_RIGHT enabled.
	9:8	virt_stop_mode	
		0	Reserved.
		1	Hard stop: VACTUAL is set to 0 on a virtual stop event.
		2	Soft stop is enabled with linear velocity ramp (from VACTUAL to v = 0).
		3	Reserved.
	10	latch_x_on_inactive_l	
		0	No latch of XACTUAL if STOPL becomes inactive.
		1	X_LATCH = XACTUAL is stored in the case STOPL becomes inactive.
	11	latch_x_on_active_l	
		0	No latch of XACTUAL if STOPL becomes active.
		1	X_LATCH = XACTUAL is stored in the case STOPL becomes active.
●→Continued on next page.			



REFERENCE_CONF 0x01 (Default value: 0x00000000)			
R/W	Bit	Val	Remarks
RW	12	latch_x_on_inactive_r	
		0	No latch of XACTUAL if STOPR becomes inactive.
		1	X_LATCH = XACTUAL is stored in the case STOPL becomes inactive.
	13	latch_x_on_active_r	
		0	No latch of XACTUAL if STOPR becomes active.
		1	X_LATCH = XACTUAL is stored in the case STOPL becomes active.
	14	stop_left_is_home	
		0	STOPL input signal is not also the HOME position.
		1	STOPL input signal is also the HOME position.
	15	stop_right_is_home	
		0	STOPR input signal is not Iso the HOME position.
		1	STOPR input signal is also the HOME position.
	19:16	home_event	
		0	Next active N event of connected ABN encoder signal indicates HOME position.
		2	HOME_REF = 1 indicates an active home event X_HOME is located at the rising edge of the active range.
		3	HOME_REF = 0 indicates negative region/position from the home position.
		4	HOME_REF = 1 indicates an active home event X_HOME is located at the falling edge of the active range.
		6	HOME_REF = 1 indicates an active home event X_HOME is located in the middle of the active range.
		9	HOME_REF = 0 indicates an active home event X_HOME is located in the middle of the active range.
		11	HOME_REF = 0 indicates an active home event X_HOME is located at the rising edge of the active range.
		12	HOME_REF = 1 indicates negative region/position from the home position.
		13	HOME_REF = 0 indicates an active home event X_HOME is located at the falling edge of the active range.
	20	start_home_tracking	
		0	No storage to X_HOME by passing home position.
		1	Storage of XACTUAL as X_HOME at next regular home event. This switch is reset after an executed homing. An XLATCH_DONE event will be also released then.
	21	clr_pos_at_target	
		0	Ramp stops at XTARGET if positioning mode is active.
		1	Set XACTUAL = 0 after XTARGET has been reached. The next ramp starts immediately.
	22	circular_movement_en	
		0	Range of XACTUAL is not limited: $-2^{31} \leq XACTUAL \leq 2^{31}-1$
		1	Range of XACTUAL is limited by X_RANGE: $-X_RANGE \leq XACTUAL \leq X_RANGE -1$
•→ Continued on next page.			



REFERENCE_CONF 0x01 (Default value: 0x00000000)			
R/W	Bit	Val	Remarks
RW	24:23		<i>pos_comp_output</i>
		0	TARGET_REACHED is set active on TARGET_REACHED event.
		1	Reserved.
		2	Reserved.
		3	TARGET_REACHED triggers on POSCOMP_REACHED event.
	25		<i>pos_comp_source</i>
		0	POS_COMP is compared to internal position XACTUAL.
		1	POS_COMP is compared with external position ENC_POS.
	26		<i>stop_on_stall</i>
		0	SPI and S/D output interface remain active in case of an stall event.
		1	SPI and S/D output interface stops motion in case of an stall event (hard stop).
	27		<i>drv_after_stall</i>
		0	No further motion in case of an active Stop-on-Stall event.
		1	Motion is possible in case of an active Stop-on-Stall event and after the Stop-on-Stall event is reset.
	29:28		<i>modified_pos_compare:</i> <i>POS_COMP_REACHED_F / event is based on comparison between XACTUAL resp. ENC_POS and</i>
		0	POS_COMP
		1	X_HOME
		2	X_LATCH resp. ENC_LATCH
		3	REV_CNT
	30		<i>automatic_cover</i>
		0	SPI output interface will not transfer automatically any cover datagram.
		1	SPI output interface sends automatically cover datagrams when VACTUAL crosses SPI_SWITCH_VEL.
	31		<i>circular_enc_en</i>
		0	Range of ENC_POS is not limited: $-2^{31} \leq ENC_POS \leq 2^{31}-1$
		1	Range of ENC_POS is limited by X_RANGE: $-X_RANGE \leq ENC_POS \leq X_RANGE - 1$

Table 61: Reference Switch Configuration 0x01



17.3. Start Switch Configuration Register START_CONF 0x02

START_CONF 0x02 (Default value: 0x00000000)			
R/W	Bit	Val	Remarks
RW	4:0	<i>start_en</i>	
		xxxx1	Alteration of <i>XTARGET</i> value requires distinct start signal.
		xxx1x	Alteration of <i>VMAX</i> value requires distinct start signal.
		xx1xx	Alteration of <i>RAMPMODE</i> value requires distinct start signal.
		x1xxx	Alteration of <i>GEAR_RATIO</i> value requires distinct start signal.
		1xxxx	Shadow Register Feature Set is enabled.
	8:5	<i>trigger_events</i>	
		0000	Timing feature set is disabled because start signal generation is disabled.
		xxx0	START pin is assigned as output.
		xxx1	External start signal is enabled as timer trigger. START pin is assigned as input.
		xx1x	<i>TARGET_REACHED</i> event is assigned as start signal trigger.
		x1xx	<i>VELOCITY_REACHED</i> event is assigned as start signal trigger.
		1xxx	<i>POSCOMP_REACHED</i> event is assigned as start signal trigger.
	9	<i>pol_start_signal</i>	
		0	START pin is low active (input resp. output).
		1	START pin is high active (input resp. output).
	10	<i>immediate_start_in</i>	
		0	Active START input signal starts internal start timer.
		1	Active START input signal is executed immediately.
	11	<i>busy_state_en</i>	
		0	START pin is only assigned as input or output.
		1	Busy start state is enabled. START pin is assigned as input with a weakly driven active start polarity or as output with a strongly driven inactive start polarity.
	15:12	<i>pipeline_en</i>	
		0000	No pipelining is active.
		xxx1	<i>X_TARGET</i> is considered for pipelining.
		xx1x	<i>POS_COMP</i> is considered for pipelining.
		x1xx	<i>GEAR_RATIO</i> is considered for pipelining.
	17:16	<i>shadow_option</i>	
		0	Single-level shadow registers for 13 relevant ramp parameters.
		1	Double-stage shadow registers for S-shaped ramps.
		2	Double-stage shadow registers for trapezoidal ramps (excl. <i>VSTOP</i>).
		3	Double-stage shadow registers for trapezoidal ramps (excl. <i>VSTART</i>).
	•→ Continued on next page.		



START_CONF 0x02 (Default value: 0x00000000)			
R/W	Bit	Val	Remarks
RW	18		<i>cyclic_shadow_regs</i>
		0	Current ramp parameters are not written back to the shadow register.
		1	Current ramp parameters are written back to the appropriate shadow register.
	19		Reserved. Set to 0.
	23:20		<i>SHADOW_MISS_CNT</i>
		U	Number of unused start internal start signals between two consecutive shadow register transfers.
	31:24		<i>XPIPE_REWRITE_REG</i>
			Current assigned pipeline registers – <i>START_CONF</i> (15:12) – are written back to <i>X_PIPEx</i> in the case of an internal start signal generation and if assigned in this register with a '1': <i>XPIPE_REWRITE_REG</i> (0) → <i>X_PIPE0</i> <i>XPIPE_REWRITE_REG</i> (1) → <i>X_PIPE1</i> <i>XPIPE_REWRITE_REG</i> (2) → <i>X_PIPE2</i> <i>XPIPE_REWRITE_REG</i> (3) → <i>X_PIPE3</i> <i>XPIPE_REWRITE_REG</i> (4) → <i>X_PIPE4</i> <i>XPIPE_REWRITE_REG</i> (5) → <i>X_PIPE5</i> <i>XPIPE_REWRITE_REG</i> (6) → <i>X_PIPE6</i> <i>XPIPE_REWRITE_REG</i> (7) → <i>X_PIPE7</i>
			Ex.: <i>START_CONF</i> (15:12) = b'0011. <i>START_CONF</i> (31:24) = b'01000010. If an internal start signal is generated, the value of <i>X_TARGET</i> is written back to <i>X_PIPE1</i> , whereas the value of <i>POS_COMP</i> is written back to <i>X_PIPE6</i> .

Table 62: Start Switch Configuration *START_CONF* 0x02

17.4. Input Filter Configuration Register INPUT_FILT_CONF 0x03

INPUT_FILT_CONF 0x03 (Default value: 0x00000000)			
R/W	Bit	Val	Remarks
RW	2:0	SR_ENC_IN	
		U	Input sample rate = $f_{clk} / 2^{SR_ENC_IN}$ for the following pins: A_SCLK, ANEG_NSCLK, B_SDI, BNEG_NSIDI, N, NNEG
	3	Reserved. Set to 0.	
	6:4	FILT_L_ENC_IN	
		U	Filter length for these pins: A_SCLK, ANEG_NSCLK, B_SDI, BNEG_NSIDI, N, NNEG. Number of sample input bits that must have equal voltage levels to provide a valid input bit.
	7	SD_FILT0	
		0	S/D input pins (STPIN/DIRIN) are not assigned to the ENC_IN input filter group.
	7	1	
		S/D input pins (STPIN/DIRIN) are also assigned to the ENC_IN input filter group.	
	10:8	SR_REF	
		U	Input sample rate = $f_{clk} / 2^{REF}$ for the following pins: STOPL, HOME_REF, STOPL
	11	Reserved. Set to 0.	
	14:12	FILT_L_ENC_IN	
		U	Filter length for the following pins: STOPL, HOME_REF, STOPL. Number of sample input bits that must have equal voltage levels to provide a valid input bit.
	15	SD_FILT1	
		0	S/D input pins (STPIN/DIRIN) are not assigned to the REF input filter group.
	15	1	
		S/D input pins (STPIN/DIRIN) are also assigned to the REF input filter group.	
	18:16	SR_S	
		U	Input sample rate = $f_{clk} / 2^S$ for the START pin.
	19	Reserved. Set to 0.	
	22:20	FILT_L_S	
		U	Filter length for the START pin. Number of sample input bits that must have equal voltage levels to provide a valid input bit.
	23	SD_FILT2	
		0	S/D input pins (STPIN/DIRIN) are not assigned to the S input filter group.
	23	1	
		S/D input pins (STPIN/DIRIN) are also assigned to the S input filter group.	
	26:24	SR_ENC_OUT	
		U	Input sample rate = $f_{clk} / 2^{SR_ENC_OUT}$ for these pins: SDODRV_SCLK, SDIDRV_NSCLK
	27	Reserved. Set to 0.	
	30:28	FILT_L_ENC_OUT	
		U	Filter length for the following pins: SDODRV_SCLK, SDIDRV_NSCLK. Number of sample input bits that must have equal voltage levels to provide a valid input bit.
	31	SD_FILT3	
		0	S/D input pins (STPIN/DIRIN) are not assigned to the ENC_OUT input filter group.
	31	1	
		S/D input pins (STPIN/DIRIN) are assigned to the ENC_OUT input filter group.	

Table 63: Input Filter Configuration Register INPUT_FILT_CONF 0x03



17.5. SPI Output Configuration Register SPI_OUT_CONF 0x04

SPI_OUT_CONF 0x04 (Default value: 0x00000000)			
R/W	Bit	Val	Remarks
RW	3:0	<i>spi_output_format</i>	
		0	SPI output interface is off .
		1	SPI output interface is connected with a SPI-DAC . SPI output values are mapped to full amplitude: Current=0 → VCC/2 Current=-max → 0 Current=max → VCC
		2	SPI output interface is connected with a SPI-DAC . SPI output values are absolute values . Phase of coilA is forwarded via STPOUT, whereas phase of coilB is forwarded via DIROUT. Phase bit = 0: positive value.
		3	SPI output interface is connected with a SPI-DAC . SPI output values are absolute values . Phase of coilA is forwarded via STPOUT, whereas phase of coilB is forwarded via DIROUT. Phase bit = 0: negative value.
		4	The actual unsigned scaling factor is forwarded via SPI output interface.
		5	Both actual signed current values CURRENTA and CURRENTB are forwarded in one datagram via SPI output interface.
		6	SPI output interface is connected with a SPI-DAC . The actual unsigned scaling factor is merged with DAC_ADDR_A value to an output datagram.
		8	SPI output interface is connected with a TMC23x stepper motor driver.
		9	SPI output interface is connected with a TMC24x stepper motor driver.
		10	SPI output interface is connected with a TMC26x/389 stepper motor driver. Configuration and current data are transferred to the stepper motor driver.
		11	SPI output interface is connected with a TMC26x stepper motor driver. Only configuration data is transferred. S/D output interface provides steps.
		12	SPI output interface is connected with a TMC2130 stepper motor driver. Only configuration data is transferred to the stepper motor driver. S/D output interface provides steps.
		13	SPI output interface is connected with a TMC2130 stepper motor driver. Configuration and current data are transferred to the stepper motor driver.
		15	Only cover datagrams are transferred via SPI output interface.
	19:13	<i>COVER_DATA_LENGTH</i>	
		U	Number of bits for the complete datagram length. Maximum value = 64 Set to 0 in case a TMC stepper motor driver is selected. The datagram length is then selected automatically.
	23:20	<i>SPI_OUT_LOW_TIME</i>	
		U	Number of clock cycles the SPI output clock remains at low level.
	27:24	<i>SPI_OUT_HIGH_TIME</i>	
		U	Number of clock cycles the SPI output clock remains at high level.
	31:28	<i>SPI_OUT_BLOCK_TIME</i>	
		U	Number of clock cycles. The NSCSDRV output remains high (inactive) after SPI output transmission.
	•→ Continued on next page.		



SPI_OUT_CONF 0x04 (Default value: 0x00000000)			
R/W	Bit	Val	Remarks
RW	5:4	<i>mixed_decay</i> (TMC23x/24x only)	
		0	Both mixed decay bits are always off.
		1	Mixed decay bits are on during falling ramps until reaching a current value of 0.
		2	Mixed decay bits are always on, except during standstill.
		3	Mixed decay bits are always on.
	6	<i>stdby_on_stall_for_24x</i> (TMC24x only)	
		0	No standby datagram is sent.
		1	In case of a Stop-on-Stall event, a standby datagram is sent to the TMC24x.
	7	<i>stall_flag_instead_of_uv_en</i> (TMC24x only)	
		0	Undervoltage flag of TMC24x is mapped at STATUS(24).
		1	Calculated stall status of TMC24x is forwarded at STATUS(24).
	10:8	STALL_LOAD_LIMIT (TMC24x only)	
		U	A stall is detected if the stall limit value STALL_LOAD_LIMIT is higher than the combination of the load bits (LD2&LD1&LD0).
	11	<i>pwm_phase_shft_en</i> (TMC24x only)	
		0	No phase shift during PWM mode.
		1	During PWM mode, the internal SinLUT microstep position MSCNT is shifted to MS_OFFSET microsteps. Consequently, the sine/cosine values have a phase shift of (MS_OFFSET / 1024 · 360°)
	4	<i>three_phase_stepper_en</i> (TMC389 only)	
		0	A 2-phase stepper motor driver is connected to the SPI output (TMC26x).
		1	A 3-phase stepper motor driver is connected to the SPI output (TMC389).
	5	<i>scale_val_transfer_en</i> (TMC26x/2130 in SD mode only)	
		0	No transfer of scale values.
		1	Transmission of current scale values to the appropriate driver registers.
	6	<i>disable_polling</i> (TMC26x/2130 in SD mode only)	
		0	Permanent transfer of polling datagrams to check driver status.
		1	No transfer of polling datagrams.
	11:7	POLL_BLOCK_MULT (TMC26x/2130 in SD mode only)	
		U	Multiplier for calculating the time interval between two consecutive polling datagrams: $t_{POLL} = (POLL_BLOCK_MULT+1) \cdot SPI_OUT_BLOCK_TIME / f_{CLK}$
	4	<i>sck_low_before_csn</i> (No TMC driver)	
		0	NSCSDRV_SDO is tied low before SCKDRV_NSDO to initiate a new data transfer.
		1	SCKDRV_NSDO is tied low before NSCSDRV_SDO to initiate a new data transfer.
	5	<i>new_out_bit_at_rise</i> (No TMC driver)	
		0	New value bit at SDODRV_SCLK is assigned at falling edge of SCKDRV_NSDO.
		1	New value bit at SDODRV_SCLK is assigned at rising edge of SCKDRV_NSDO.
•→ Continued on next page.			



SPI_OUT_CONF 0x04 (Default value: 0x00000000)			
R/W	Bit	Val	Remarks
	11:7		<i>DAC_CMD_LENGTH</i> (SPI-DAC only)
		U	Number of bits for command address if one of the SPI-DAC options is selected
RW	12		Reserved. Set to 0.
	23:4		<i>SSI_OUT_MTIME</i> (Serial encoder output only)
		U	Monoflop time for SSI output interface: Delay time [clock cycles] during which the absolute encoder data remain stable after the last master request.

Table 64: SPI Output Configuration Register SPI_OUT_CONF 0x04



17.6. Current Scaling Configuration Register CURRENT_CONF 0x05

CURRENT_CONF 0x05 (Default: 0x00000000)			
R/W	Bit	Val	Remarks
RW	0	<i>hold_current_scale_en</i>	
		0	No hold current scaling during standby phase.
		1	Hold current scaling during standby phase.
	1	<i>drive_current_scale_en</i>	
		0	No drive current scaling during motion.
		1	Drive current scaling during motion.
	2	<i>boost_current_on_acc_en</i>	
		0	No boost current scaling for deceleration ramps.
		1	Boost current scaling if <i>RAMP_STATE</i> = b'01 (acceleration slopes).
	3	<i>boost_current_on_dec_en</i>	
		0	No boost current scaling for deceleration ramps.
		1	Boost current scaling if <i>RAMP_STATE</i> = b'10 (deceleration slopes).
	4	<i>boost_current_after_start_en</i>	
		0	No boost current at ramp start.
		1	Temporary boost current if <i>VACTUAL</i> = 0 and new ramp starts.
	5	<i>sec_drive_current_scale_en</i>	
		0	One drive current value for the whole motion ramp.
		1	Second drive current scaling for <i>VACTUAL</i> > <i>VDRV_SCALE_LIMIT</i> .
	6	<i>freewheeling_en</i>	
		0	No freewheeling.
		1	Freewheeling after standby phase.
	7	<i>closed_loop_scale_en</i>	
		0	No closed-loop current scaling.
		1	Closed-loop current scaling – <i>CURRENT_CONF</i> (6:0) = 0 is set automatically ! Turn off for closed-loop calibration with maximum current!
	8	<i>pwm_scale_en</i>	
		0	PWM scaling is disabled.
		1	PWM scaling is enabled.
	15:9	Reserved. Set to 0x00.	
	31:16	<i>PWM_AMPL</i>	
		U	PWM amplitude during Voltage PWM mode at <i>VACTUAL</i> = 0.
			i Maximum duty cycle = $(0.5 + (PWM_AMPL + 1) / 2^{17})$ Minimum duty cycle = $(0.5 - (PWM_AMPL + 1) / 2^{17})$ PWM_AMPL = $2^{16} - 1$ at <i>VACTUAL</i> = <i>PWM_VMAX</i>

Table 65: Current Scale Configuration (0x05)



17.7. Current Scale Values Register **SCALE_VALUES 0x06**

SCALE_VALUES 0x06 (Default: 0xFFFFFFFF)				
R/W	Bit	Val	Scaling Value Name	Remarks
RW	7:0	U	BOOST_SCALE_VAL	Open-loop boost scaling value.
			CL_IMIN	Closed-loop minimum scaling value.
	15:8	U	DRV1_SCALE_VAL	Open-loop first drive scaling value.
			CL_IMAX	Closed-loop maximum scaling value.
	23:16	U	DRV2_SCALE_VAL	Open-loop second drive scaling value.
			CL_START_UP	ENC_POS_DEV value at which closed-loop scaling increases the current scaling value above CL_IMIN.
	31:24	U	HOLD_SCALE_VAL	Open-loop standby scaling value.
			CL_START_DOWN	ENC_POS_DEV value at which closed-loop scaling decreases the current scaling value below CL_IMAX. i Recommended: Set to 0 to automatically assign to CL_BETA.

Table 66: Current Scale Values (0x06)

NOTE:

- BOOST_SCALE_VAL, DRV1/DRV2_SCALE_VAL, HOLD_SCALE_VAL, CL_IMIN, CL_IMAX.
- Real scaling value = $(x+1) / 32$ if spi_output_format = b'1011 or b'1100.
- = $(x+1) / 256$ any other spi_output_format setting.



17.8. Encoder Signal Configuration (0x07)

ENC_IN_CONF 0x07 (Default 0x00000400)			
R/W	Bit	Val	Description
RW	0	enc_sel_decimal	
		0	Encoder constant represents a binary number.
		1	Encoder constant represents a decimal number (for ABN only).
	1	clear_on_n	
		0	ENC_POS is not set to ENC_RESET_VAL.
		1	ENC_POS is set to ENC_RESET_VAL on every N event in case clr_latch_cont_on_n=1, or on the next N event in case clr_latch_once_on_n=1. ! Do NOT use during closed-loop operation!
	2	clr_latch_cont_on_n	
		0	Value of ENC_POS is not cleared and/or latched on every N event.
		1	Value of ENC_POS is cleared and/or latched on every N event.
	3	clr_latch_once_on_n	
		0	Value of ENC_POS is not cleared and/or latched on the next N event.
		1	Value of ENC_POS is cleared and/or latched on the next N event. i This bit is set to 0 after latching/clearing once.
	4	pol_n	
		0	Active polarity for N event is low active.
		1	Active polarity for N event is high active.
	6:5	n_chan_sensitivity	
		0	N event is active as long as N equals active N event polarity.
		1	N event triggers when N switches to active N event polarity.
		2	N event triggers when N switches to inactive N event polarity.
	7	pol_a_for_n	
		0	A polarity has to be low for a valid N event.
		1	A polarity has to be high for a valid N event.
	8	pol_b_for_n	
		0	B polarity has to be low for valid N event
		1	B polarity has to be high for valid N event
	9	ignore_ab	
		0	TMC4361 considers A and B polarities for valid N event.
		1	Polarities of A and B signals for a valid N event are ignored.

•→Continued on next page.



ENC_IN_CONF 0x07 (Default 0x00000400)			
R/W	Bit	Val	Description
RW	10	<i>latch_enc_on_n</i>	
		0	ENC_POS is not latched.
		1	ENC_POS is latched to ENC_LATCH on every N event in case <i>clr_latch_cont_on_n</i> =1, or on the next N event in case <i>clr_latch_once_on_n</i> =1.
	11	<i>latch_x_on_n</i>	
		0	XACTUAL is not latched.
		1	XACTUAL is latched to X_LATCH on every N event in case <i>clr_latch_cont_on_n</i> =1, or on the next N event in case <i>clr_latch_once_on_n</i> =1.
	12	<i>multi_turn_in_en</i> (Absolute encoder only)	
		0	Connected serial encoder transmits singleturn values.
		1	Connected serial encoder input transmits singleturn and multiturn values.
	13	<i>multi_turn_in_signed</i> (Absolute encoder only)	
		0	Multiturn values from serial encoder input are unsigned numbers.
		1	Multiturn values from serial encoder input are signed numbers.
	14	<i>multi_turn_out_en</i> (Serial encoder output only)	
		0	Serial encoder output transmits singleturn values.
		1	Serial encoder output transmits singleturn and multiturn values.
	15	<i>use_usteps_instead_of_xrange</i>	
		0	X_RANGE is valid in case circular motion is also enabled for encoders.
		1	USTEPS_PER_REV is valid in case circular motion is also enabled for encoders.
	16	<i>calc_multi_turn_behav</i> (Absolute encoder only)	
		0	No multiturn calculation.
		1	TMC4361 calculates internally multiturn data for singleturn encoder data.
	17	<i>ssi_multi_cycle_data</i> (Absolute encoder only)	
		0	Every SSI value request is executed once.
		1	Every SSI value request is executed twice.
	18	<i>ssi_gray_code_en</i> (Absolute encoder only)	
		0	SSI input data is binary-coded.
		1	SSI input data is gray-coded.
	19	<i>left_aligned_data</i> (Absolute encoder only)	
		0	Serial input data is aligned right (first flags, then data).
		1	Serial input data is aligned left (first data, then flags).
	•→Turn page.		



ENC_IN_CONF 0x07 (Default 0x00000400)			
R/W	Bit	Val	Description
RW	20	spi_data_on_cs (SPI encoder only)	
		0	BNEG_NSUDI provides serial output data at next serial clock line (A_SCLK) transition.
		1	BNEG_NSUDI provides serial output data immediately in case negated chip select line ANEG_NSCLK switches to low level.
	21	spi_low_before_cs (SPI encoder only)	
		0	Serial clock line A_SCLK switches to low level after negated chip select line ANEG_NSCLK switches to low level.
		1	Serial clock line A_SCLK switches to low level before negated chip select line ANEG_NSCLK switches to low level.
	23:22	regulation_modus	
		0	No internal regulation on encoder feedback data.
		1	Closed-loop operation is enabled. ! Use full microstep resolution only! (256 μSteps/FS → MSTEPS_PER_FS=0)
		2	PID regulation is enabled. Pulse generator base velocity equals 0.
		3	PID regulation is enabled. Pulse generator base velocity equals VACTUAL.
	24	cl_calibration_en (Closed-loop operation only)	
		0	Closed-loop calibration is deactivated.
		1	Closed-loop calibration is active. ! Use maximum current without scaling during calibration. ! It is recommend to keep the motor driver at fullstep position with no motion occurrence during the calibration process.
	25	cl_emf_en (Closed-loop operation only)	
		0	Back-EMF compensation deactivated during closed-loop operation.
		1	Back-EMF compensation is enabled during closed-loop operation. Closed-loop operation compensates Back-EMF in case VACTUAL > CL_VMIN.
	26	cl_clr_xact (Closed-loop operation only)	
		0	XACTUAL is not reset to ENC_POS during closed-loop operation.
		1	XACTUAL is set to ENC_POS in case ENC_POS_DEV > ENC_POS_DEV_TOL during closed-loop operation. ! This feature must only be used if understood completely.
	27	cl_vlimit_en (Closed-loop operation only)	
		0	No catch-up velocity limit during closed-loop regulation.
		1	Catch-up velocity during closed-loop operation is limited by internal PI regulator.
	28	cl_velocity_mode_en (Closed-loop operation only)	
		0	Closed-loop velocity mode is deactivated.
		1	Closed-loop velocity mode is deactivated. In case ENC_POS_DEV > 768, XACTUAL is adjusted accordingly.
	29	invert_enc_dir	
		0	Encoder direction is NOT inverted internally.
		1	Encoder direction is inverted internally.
•→Turn page.			



ENC_IN_CONF 0x07 (Default 0x00000400)			
R/W	Bit	Val	Description
RW	30		<i>enc_out_gray</i> (Serial encoder output only)
		0	SSI output data is binary-encoded.
		1	SSI output data is gray-encoded.
	31		<i>no_enc_vel_preproc</i> (Incremental ABN encoder)
		0	AB signal is preprocessed for internal encoder velocity calculation.
		1	No AB signal preprocessing. ! It is recommend to maintain AB preprocessing in order to filter encoder resonances.
			<i>serial_enc_variation_limit</i> (Absolute encoder)
		0	No variation limit on absolute encoder data.
		1	Two consecutive serial encoder values must no deviate from specified limit to be valid. In case $ ENC_POS_x - ENC_POS_{x-1} > 1/8 \cdot SER_ENC_VARIATION \cdot ENC_IN_RES$, ENC_POS_x is not valid and is not assigned to ENC_POS .

Table 67: Encoder Signal Configuration ENC_IN_CONF (0x07)

17.9. Serial Encoder Data Input Configuration (0x08)

ENC_IN_DATA 0x08 (Default: 0x00000000)			
R/W	Bit	Val	Remarks
RW	4:0		<i>SINGLE_TURN_RES</i> (Default: 0x00)
		U	Number of angle data bits within one revolution = $SINGLE_TURN_RES + 1$
	9:5		<i>MULTI_TURN_RES</i> (Default: 0x00)
		U	Number of data bits for revolution count = $MULTI_TURN_RES + 1$
	11:10		<i>STATUS_BIT_CNT</i> (Default: 0x0)
		U	Number of status data bits
	15:12		Reserved. Set to 0x0.
	23:16		<i>SERIAL_ADDR_BITS</i> (Default: 0x00) (SPI encoder only)
		U	Number of address bits within one SPI datagram for SPI encoder configuration
	31:24		<i>SERIAL_DATA_BITS</i> (Default: 0x00) (SPI encoder only)
		U	Number of data bits within one SPI datagram for SPI encoder configuration

Table 68: Serial Encoder Data Input Configuration ENC_IN_DATA (0x08)



17.10. Serial Encoder Data Output Configuration (0x09)

ENC_OUT_DATA 0x09 (Default: 0x00000000)			
R/W	Bit	Val	Remarks
RW	4:0	<i>SINGLE_TURN_RES_OUT</i> (Default: 0x00)	
		U	Number of angle data bits within one revolution = <i>SINGLE_TURN_RES_OUT</i> + 1
	9:5	<i>MULTI_TURN_RES_OUT</i> (Default: 0x00)	
		U	Number of data bits for revolution count = <i>MULTI_TURN_RES_OUT</i> + 1
	31:12	Reserved. Set to 0x00000.	

Table 69: Serial Encoder Data Output Configuration ENC_OUT_DATA (0x09)



17.11. Motor Driver Settings Register STEP_CONF 0x0A

STEP_CONF 0x0A (Default: 0x00FB0C80)			
R/W	Bit	Val	Remarks
RW	3:0	<i>MSTEP_PER_FS</i> (Default: 0x0)	
		0	Highest microsteps resolution: 256 microsteps per fullstep. i Set to 256 for closed-loop operation. i When using a Step/Dir driver, it must be capable of a 256 resolution via Step/Dir input for best performance (but lower resolution Step/Dir drivers can be used as well).
		1	128 microsteps per fullstep.
		2	64 microsteps per fullstep.
		3	32 microsteps per fullstep.
		4	16 microsteps per fullstep.
		5	8 microsteps per fullstep.
		6	4 microsteps per fullstep.
		7	Halfsteps: 2 microsteps per fullstep.
		8	Full steps (maximum possible setting)
	15:4	<i>FS_PER_REV</i> (Default: 0x0C8)	
		U	Fullsteps per motor axis revolution
	23:16	<i>MSTATUS_SELECTION</i> (Default: 0xFB)	
			Selection of motor driver status bits for SPI response datagrams: ORed with Motor Driver Status Register Set (7:0): if set here and a particular flag is set from the motor stepper driver, an event will be generated at <i>EVENTS</i> (30)
	31:24	Reserved. Set to 0x00.	

Table 70: Motor Driver Settings (0x0A)



17.12. Event Selection Registers 0x0B..0x0D

Event Selection Registers			
R/W	Addr	Bit	Remarks
RW	0x0B	<i>SPI_STATUS_SELECTION</i> (Default: 0x82029805)	
		31:0	Events selection for SPI datagrams: Event bits of <i>EVENTS</i> register 0x0E that are selected (=1) in this register are forwarded to the eight status bits that are transferred with every SPI datagram (first eight bits from LSB are significant!).
	0x0C	<i>EVENT_CLEAR_CONF</i> (Default: 0x00000000)	
		31:0	Event protection configuration: Event bits of <i>EVENTS</i> register 0x0E that are selected in this register (=1) are not cleared during the readout process of <i>EVENTS</i> register 0x0E.
	0x0D	<i>INTR_CONF</i> (Default: 0x00000000)	
		31:0	Event selection for INTR output: All Event bits of <i>EVENTS</i> register 0x0E that are selected here (=1) are ORed with interrupt event register set: if any of the selected events is active, an interrupt at INTR is generated.

Table 71: Event Selection Registers 0x0B...0x0D



17.13. Status Event Register (0x0E)

Status Event Register <i>EVENTS</i> 0x0E		
R/W	Bit	Description
R+C	0	<i>TARGET_REACHED</i> has been triggered.
	1	<i>POS_COMP_REACHED</i> has been triggered.
	2	<i>VEL_REACHED</i> has been triggered.
	3	<i>VEL_STATE</i> = b'00 has been triggered (<i>VACTUAL</i> = 0).
	4	<i>VEL_STATE</i> = b'01 has been triggered (<i>VACTUAL</i> > 0).
	5	<i>VEL_STATE</i> = b'10 has been triggered (<i>VACTUAL</i> < 0).
	6	<i>RAMP_STATE</i> = b'00 has been triggered (<i>AActual</i> = 0, <i>VACTUAL</i> is constant).
	7	<i>RAMP_STATE</i> = b'01 has been triggered (<i> VACTUAL </i> increases).
	8	<i>RAMP_STATE</i> = b'10 has been triggered (<i> VACTUAL </i> increases).
	9	<i>MAX_PHASE_TRAP</i> : Trapezoidal ramp has reached its limit speed using maximum values for <i>AMAX</i> or <i>DMAX</i> (<i> VACTUAL </i> > <i>VBREAK</i> ; <i>VBREAK</i> ≠ 0).
	10	<i>FROZEN</i> : <i>NFREEZE</i> has switched to low level. i Reset TMC4361 for further motion.
	11	<i>STOPL</i> has been triggered. Motion in negative direction is not executed until this event is cleared and (<i>STOPL</i> is not active any more or <i>stop_left_en</i> is set to 0).
	12	<i>STOPR</i> has been triggered. Motion in positive direction is not executed until this event is cleared and (<i>STOPR</i> is not active any more or <i>stop_right_en</i> is set to 0).
	13	<i>VSTOPL_ACTIVE</i> : <i>VSTOPL</i> has been activated. No further motion in negative direction until this event is cleared and (a new value is chosen for <i>VSTOPL</i> or <i>virtual_left_limit_en</i> is set to 0).
	14	<i>VSTOPR_ACTIVE</i> : <i>VSTOPR</i> has been activated. No further motion in positive direction until this event is cleared and (a new value is chosen for <i>VSTOPR</i> or <i>virtual_right_limit_en</i> is set to 0).
	15	<i>HOME_ERROR</i> : Unmatched <i>HOME_REF</i> polarity and <i>HOME</i> is outside of safety margin.
	16	<i>XLATCH_DONE</i> indicates if <i>X_LATCH</i> was rewritten or homing process has been completed.
	17	<i>FS_ACTIVE</i> : Fullstep motion has been activated.
	18	<i>ENC_FAIL</i> : Mismatch between <i>XACTUAL</i> and <i>ENC_POS</i> has exceeded specified limit.
	19	<i>N_ACTIVE</i> : <i>N</i> event has been activated.
	20	<i>ENC_DONE</i> indicates if <i>ENC_LATCH</i> was rewritten.
	21	<i>SER_ENC_DATA_FAIL</i> : Failure during multi-cycle data evaluation or between two consecutive data requests has occurred.
	22	Reserved.
	23	<i>SER_DATA_DONE</i> : Configuration data was received from serial SPI encoder.
	24	One of the <i>SERIAL_ENC_Flags</i> was set.
	25	<i>COVER_DONE</i> : SPI datagram was sent to the motor driver.
	26	<i>ENC_VELO</i> : Encoder velocity has reached 0.
	27	<i>CL_MAX</i> : Closed-loop commutation angle has reached maximum value.
	28	<i>CL_FIT</i> : Closed-loop deviation has reached inner limit.
	29	<i>STOP_ON_STALL</i> : Motor stall detected. Motor ramp has stopped.
	30	<i>MOTOR_EV</i> : One of the selected TMC motor driver flags was triggered.
	31	Reset was triggered.

Table 72: Status Event Register *EVENTS* (0x0E)



17.14. Status Flag Register (0x0F)

Status Flag Register <i>STATUS</i> 0x0F		
R/W	Bit	Description
R	0	<i>TARGET_REACHED_F</i> is set high if <i>XACTUAL</i> = <i>XTARGET</i>
	1	<i>POS_COMP_REACHED_F</i> is set high if <i>XACTUAL</i> = <i>POS_COMP</i>
	2	<i>VEL_REACHED_F</i> is set high if <i>VACTUAL</i> = <i>VMAX</i>
	4:3	<i>VEL_STATE_F</i> : Current velocity state: 0 → <i>VACTUAL</i> = 0; 1 → <i>VACTUAL</i> > 0; 2 → <i>VACTUAL</i> < 0
	6:5	<i>RAMP_STATE_F</i> : Current ramp state: 0 → <i>AACTUAL</i> = 0; 1 → <i>AACTUAL</i> increases (acceleration); 2 → <i>AACTUAL</i> decreases (deceleration)
	7	<i>STOPL_ACTIVE_F</i> : Left stop switch is active.
	8	<i>STOPR_ACTIVE_F</i> : Right stop switch is active.
	9	<i>VSTOPL_ACTIVE_F</i> : Left virtual stop switch is active.
	10	<i>VSTOPR_ACTIVE_F</i> : Right virtual stop switch is active.
	11	<i>ACTIVE_STALL_F</i> : Motor stall is detected and <i>VACTUAL</i> > <i>VSTALL_LIMIT</i> .
	12	<i>HOME_ERROR_F</i> : HOME_REF input signal level is not equal to expected home level.
	13	<i>FS_ACTIVE_F</i> : Fullstep operation is active.
	14	<i>ENC_FAIL_F</i> : Mismatch between <i>XACTUAL</i> and <i>ENC_POS</i> is out of tolerated range.
	15	<i>N_ACTIVE_F</i> : N event is active.
	16	<i>ENC_LATCH_F</i> : <i>ENC_LATCH</i> is rewritten.
	17	Applies to absolute encoders only: <i>MULTI_CYCLE_FAIL_F</i> indicates a failure during last multi cycle data evaluation.
		Applies to absolute encoders only: <i>SER_ENC_VAR_F</i> indicates a failure during last serial data evaluation due to a substantial deviation between two consecutive serial data values.
	18	Reserved.
	19	<i>CL_FIT_F</i> : Active if <i>ENC_POS_DEV</i> < <i>CL_TOLERANCE</i> . The current mismatch between <i>XACTUAL</i> and <i>ENC_POS</i> is within tolerated range.
	23:20	Applies to absolute encoders only: <i>SERIAL_ENC_FLAGS</i> received from encoder. These flags are reset with a new encoder transfer request.
	24	TMC26x / TMC2130 only: <i>SG</i> : StallGuard2 status
		Optional for TMC24x only: Calculated stallGuard status.
	25	TMC23x / TMC24x only: <i>UV_SF</i> : Undervoltage flag.
		All TMC motor drivers: <i>OT</i> : Overtemperature shutdown.
	26	All TMC motor drivers: <i>OTPW</i> : Overtemperature warning.
	27	TMC26x / TMC2130 only: <i>S2GA</i> : Short to ground detection bit for high side MOSFET of coil A.
		TMC23x / TMC24x only: <i>OCA</i> : Overcurrent bridge A.
	28	TMC26x / TMC2130 only: <i>S2GB</i> : Short to ground detection bit for high side MOSFET of coil B.
		TMC23x / TMC24x only: <i>OCB</i> : Overcurrent bridge B.
	29	All TMC motor drivers: <i>OLA</i> : Open load indicator of coil A.
	30	All TMC motor drivers: <i>OLB</i> : Open load indicator of coil B.
	31	TMC26x / TMC2130 only: <i>STST</i> : Standstill indicator.
		TMC23x / TMC24x only: <i>OCHS</i> : Overcurrent high side.

Table 73: Status Flag Register *STATUS* (0x0F)



17.15. Configuration Registers: Closed-Loop, Switches, etc.

Various Configuration Registers: Closed-loop, Switches...				
R/W	Addr	Bit	Val	Description
RW	0x10	15:0	U	<i>STP_LENGTH_ADD</i> (Default: 0x0000)
				Additional length [# clock cycles] for active step polarity to indicate an active output step at STPOUT.
	0x11	31:16	U	<i>DIR_SETUP_TIME</i> (Default: 0x0000)
				Delay [# clock cycles] between DIROUT and STPOUT voltage level changes.
	0x12	31:0	U	<i>START_OUT_ADD</i> (Default: 0x00000000)
				Additional length [# clock cycles] for active start signal. Active start signal length = 1+START_OUT_ADD
	0x13	31:0	S	<i>GEAR_RATIO</i> (Default: 0x01000000)
				Constant value that is added to the internal position counter by an active step at STPIN. Value representation: 8 digits and 24 decimal places.
	0x14	31:0	U	<i>START_DELAY</i> (Default: 0x00000000)
				Delay time [# clock cycles] between start trigger and internal start signal release.
	0x15	31:0	U	Reserved. Set to 0x00000000.
				<i>STDBY_DELAY</i> (Default: 0x00000000)
	0x16	31:0	U	Delay time [# clock cycles] between ramp stop and activating standby phase.
				<i>FREEWHEEL_DELAY</i> (Default: 0x00000000)
	0x17	23:0	U	Delay time [# clock cycles] between initialization of active standby phase and freewheeling initialization.
				<i>VDRV_SCALE_LIMIT</i> (Default: 0x00000000) (Voltage PWM mode is not active)
	0x18	23:0	U	Drive scaling separator: <i>DRV2_SCALE_VAL</i> is active in case $V_{ACTUAL} > VDRV_SCALE_LIMIT$ <i>DRV1_SCALE_VAL</i> is active in case $V_{ACTUAL} \leq VDRV_SCALE_LIMIT$
				<i>PWM_VMAX</i> (Default: 0x00000000) (Voltage PWM is enabled)
	0x19	23:0	U	PWM velocity value at which maximal scale parameter value 1.0 is reached.
				<i>UP_SCALE_DELAY</i> (Default: 0x000000) (Open-loop operation)
	0x1A	23:0	U	Increment delay [# clock cycles]. The value defines the clock cycles, which are used to increase the current scale value for one step towards higher values.
				<i>CL_UPSCALE_DELAY</i> (Default: 0x000000) (Closed-loop operation)
	0x1B	23:0	U	Increment delay [# clock cycles]. The value defines the clock cycles, which are used to increase the current scale value for one step towards higher current values during closed-loop operation.
				<i>HOLD_SCALE_DELAY</i> (Default: 0x000000) (Open-loop operation)
	0x1C	23:0	U	Decrement delay [# clock cycles] to decrease the actual scale value by one step towards hold current.
				<i>CL_DNSCALE_DELAY</i> (Default: 0x000000) (Closed-loop operation)
	0x1D	23:0	U	Decrement delay [# clock cycles] to decrease the current scale value by one step towards lower current values during closed-loop operation.
				<i>CL_DNSCALE_DELAY</i> (Default: 0x000000) (Closed-loop operation)



Various Configuration Registers: Closed-loop, Switches...				
R/W	Addr	Bit	Val	Description
RW	0x1A	23:0		<i>DRV_SCALE_DELAY</i> (Default: 0x000000)
			U	Decrement delay [# clock cycles], which signifies current scale value decrease by one step towards lower value.
	0x1B	31:0		<i>BOOST_TIME</i> (Default: 0x00000000)
			U	Time [# clk cycles] after a ramp start when boost scaling is active.
	0x1C	8:0		<i>CL_BETA</i> (0x0FF)
			U	Maximum commutation angle for closed-loop regulation. ➤ Set CL_BETA > 255 carefully (esp. if <i>cl_vlimit_en</i> = 1). - i Exactly 255 is recommended for best performance.
		23:16		<i>CL_GAMMA</i> (Default: 0xFF)
			U	Maximum balancing angle to compensate back-EMF at higher velocities during closed-loop regulation.
	0x1D	15:0		<i>DAC_ADDR_A</i> (Default: 0x0000)
			U	Fixed command/address, which is sent via SPI output before sending CURRENTA_SPI values.
		31:16		<i>DAC_ADDR_B</i> (Default: 0x0000)
			U	Fixed command/address, which is sent via SPI output before sending current CURRENTB_SPI values.
	0x1E	15:0		<i>HOME_SAFETY_MARGIN</i> (Default: 0x0000)
			U	HOME_REF polarity can be invalid within $X_{HOME} \pm HOME_SAFETY_MARGIN$, which is not flagged as error.
	0x1F	15:0		<i>PWM_FREQ</i> (Default: 0x0280) (Voltage PWM is enabled)
			U	Number of clock cycles for one PWM period.
		11:0		<i>CHOPSYNC_DIV</i> (Default: 0x0280) (ChopSync for TMC23x/24x is enabled)
			U	Chopper clock divider that defines the chopper frequency f_{OSC} : $f_{OSC} = f_{CLK} / CHOPSYNC_DIV$ with $96 \leq CHOPSYNC_DIV \leq 818$

Table 74: Various Configuration Registers: Closed-loop, Switches, etc.



17.16. Ramp Generator Registers

Ramp Generator Registers				
R/W	Addr	Bit	Val	Description
RW	0x20	RAMPMODE (Default: 0x0)		
		2	Operation Mode:	
			1	Positioning mode: <i>XTARGET</i> is superior target of velocity ramp.
		0	Velocity mode: <i>VMAX</i> is superior target of velocity ramp.	
		1:0	Motion Profile:	
			0	No ramp: <i>VACTUAL</i> follows only <i>VMAX</i> (rectangle velocity shape).
			1	Trapezoidal ramp (incl. sixPoint ramp): Consideration of acceleration and deceleration values for generating <i>VACTUAL</i> without adapting the acceleration values.
	2	S-shaped ramp: Consideration of all ramp values (incl. bow values) for generating <i>VACTUAL</i> .		
RW	0x21	31:0	<i>XACTUAL</i> (Default: 0x00000000)	
			S	Actual internal motor position [pulses]: $-2^{31} \leq XACTUAL \leq 2^{31} - 1$
R	0x22	31:0	<i>VACTUAL</i> (Default: 0x00000000)	
			S	Actual ramp generator velocity [pulses per second]: $1 \text{ pps} \leq VACTUAL \leq CLK_FREQ \cdot \frac{1}{2} \text{ pulses}$ ($f_{CLK} = 16 \text{ MHz} \rightarrow 8 \text{ Mpps}$)
R	0x23	31:0	<i>AACTUAL</i> (Default: 0x00000000)	
			S	Actual acceleration/deceleration value [pulses per sec ²]: $-2^{31} \text{ pps}^2 \leq AACTUAL \leq 2^{31} - 1$ $1 \text{ pps}^2 \leq AACTUAL $
RW	0x24	31:0	<i>VMAX</i> (Default: 0x00000000)	
			S	Maximum ramp generator velocity in positioning mode or
				Target ramp generator velocity in velocity mode and no ramp motion profile.
				Value representation: 23 digits and 8 decimal places $4 \text{ mpps} \leq VMAX [\text{pps}] \leq CLK_FREQ \cdot \frac{1}{2} \text{ pulses}$ $f_{CLK} = 16 \text{ MHz} \rightarrow VMAX \leq 8 \text{ Mpps}$
RW	0x25	30:0	<i>VSTART</i> (Default: 0x00000000)	
			U	Absolute start velocity in <i>positioning mode</i> and <i>velocity mode</i> In case <i>VSTART</i> is used: no first bow phase <i>B</i> ₁ for S-shaped ramps
				<i>VSTART</i> in positioning mode: In case <i>VACTUAL</i> = 0 and <i>XTARGET</i> ≠ <i>XACTUAL</i> : no acceleration phase for <i>VACTUAL</i> = 0 → <i>VSTART</i> .
				<i>VSTART</i> in velocity mode: In case <i>VACTUAL</i> = 0 and <i>VACTUAL</i> ≠ <i>VMAX</i> : no acceleration phase for <i>VACTUAL</i> = 0 → <i>VSTART</i> .
				Value representation: 23 digits and 8 decimal places.
•→ Continued on next page.				



Ramp Generator Registers				
R/W	Addr	Bit	Val	Description
RW	0x26	30:0		<i>VSTOP (Default: 0x00000000)</i>
			U	Absolute stop velocity in positioning mode and velocity mode. In case <i>VSTOP</i> is used: no last bow phase B_4 for S-shaped ramps. In case <i>VSTOP</i> is very small and positioning mode is used, it is possible that the ramp is finished with a constant $V_{ACTUAL} = V_{STOP}$ until <i>XTARGET</i> is reached.
				<i>VSTOP</i> in positioning mode: In case $V_{ACTUAL} \leq V_{STOP}$ and $XTARGET = X_{ACTUAL}$: V_{ACTUAL} is immediately set to 0.
				<i>VSTOP</i> in velocity mode: In case $V_{ACTUAL} \leq V_{STOP}$ and $V_{MAX} = 0$: V_{ACTUAL} is immediately set to 0.
				Value representation: 23 digits and 8 decimal places.
	0x27	30:0		<i>VBREAK (Default: 0x00000000)</i>
			U	Absolute break velocity in positioning mode and in velocity mode, This only applies for trapezoidal ramp motion profiles. In case $VBREAK = 0$: pure linear ramps are generated with $AMAX / DMAX$ only.
				In case $ V_{ACTUAL} < VBREAK$: $ A_{ACTUAL} = A_{START}$ or D_{FINAL} In case $ V_{ACTUAL} \geq VBREAK$: $ A_{ACTUAL} = AMAX$ or $DMAX$
				! Always set $VBREAK > VSTOP$! If $VBREAK \neq 0$.
				Value representation: 23 digits and 8 decimal places.
	0x28	23:0		<i>AMAX (Default: 0x000000)</i>
			U	S-shaped ramp motion profile: Maximum acceleration value.
				Trapezoidal ramp motion profile: Acceleration value in case $ V_{ACTUAL} \geq VBREAK$ or in case $VBREAK = 0$.
				Value representation: Frequency mode: [pulses per sec^2] 22 digits and 2 decimal places: $250 \text{ mpps}^2 \leq AMAX \leq 4 \text{ Mpps}^2$ Direct mode: [Δv per clk cycle] $a[\Delta v \text{ per clk_cycle}] = AMAX / 2^{37}$ $AMAX [\text{pps}^2] = AMAX / 2^{37} \cdot f_{CLK}^2$! Set $AMAX \leq 65536$.
	0x29	23:0		<i>DMAX (Default: 0x000000)</i>
			U	S-shaped ramp motion profile: Maximum deceleration value.
				Trapezoidal ramp motion profile: Deceleration value if $ V_{ACTUAL} \geq VBREAK$ or if $VBREAK = 0$.
				Value representation: Frequency mode: [pulses per sec^2] 22 digits and 2 decimal places: $250 \text{ mpps}^2 \leq DMAX \leq 4 \text{ Mpps}^2$ Direct mode: [Δv per clk cycle] $d[\Delta v \text{ per clk_cycle}] = DMAX / 2^{37}$ $DMAX [\text{pps}^2] = DMAX / 2^{37} \cdot f_{CLK}^2$! Set $DMAX \leq 65536$.



Ramp Generator Registers				
R/W	Addr	Bit	Val	Description
RW	0x2A	23:0		<i>ASTART (Default: 0x000000)</i>
			U	S-shaped ramp motion profile: start acceleration value.
				Trapezoidal ramp motion profile: Acceleration value in case $ VACTUAL < VBREAK$.
				Acceleration value after switching from external to internal step control.
				Value representation: Frequency mode: [pulses per sec ²] 22 digits and 2 decimal places: $250 \text{ mpps}^2 \leq ASTART \leq 4 \text{ Mpps}^2$ Direct mode: [Δv per clk cycle] $a[\Delta v \text{ per clk_cycle}] = ASTART / 2^{37}$ $ASTART [\text{pps}^2] = ASTART / 2^{37} \cdot f_{CLK}^2$! Set $ASTART \leq 65536$.
		31		Sign of <i>AActual</i> after switching from external to internal step control.
	0x2B	23:0		<i>DFINAL (Default: 0x000000)</i>
			U	S-shaped ramp motion profile: Stop deceleration value, which is not used during positioning mode.
				Trapezoidal ramp motion profile: Deceleration value in case $ VACTUAL < VBREAK$.
				Value representation: Frequency mode: [pulses per sec ²] 22 digits and 2 decimal places: $250 \text{ mpps}^2 \leq DFINAL \leq 4 \text{ Mpps}^2$ Direct mode: [Δv per clk cycle] $d[\Delta v \text{ per clk_cycle}] = DFINAL / 2^{37}$ $DFINAL [\text{pps}^2] = DFINAL / 2^{37} \cdot f_{CLK}^2$! Set $DFINAL \leq 65536$.
	0x2C	23	U	Deceleration value for an automatic linear stop ramp to $VACTUAL = 0$. <i>DSTOP</i> is used with activated external stop switches (STOPL or STOPR) if <i>soft_stop_enable</i> is set to 1; or with activated virtual stop switches and <i>virt_stop_mode</i> is set to 2.
				Value representation: Frequency mode: [pulses per sec ²] 22 digits and 2 decimal places: $250 \text{ mpps}^2 \leq DSTOP \leq 4 \text{ Mpps}^2$ Direct mode: [Δv per clk cycle] $d[\Delta v \text{ per clk_cycle}] = DSTOP / 2^{37}$ $DSTOP [\text{pps}^2] = DSTOP / 2^{37} \cdot f_{CLK}^2$! Set $DSTOP \leq 65536$.
•→ Continued on next page.				



Ramp Generator Registers				
R/W	Addr	Bit	Val	Description
RW	0x2D	23:0		<i>BOW1 (Default: 0x000000)</i>
			U	Bow value 1 (first bow B ₁ of the acceleration ramp).
				Value representation: Frequency mode: [pulses per sec ³] 24 digits and 0 decimal places: $1 \text{ pps}^3 \leq BOW1 \leq 16 \text{ Mpps}^3$ Direct mode: [Δa per clk cycle] $\text{bow}[\text{av per clk_cycle}] = BOW1 / 2^{53}$ $BOW1 [\text{pps}^3] = BOW1 / 2^{53} \cdot f_{\text{CLK}}^3$! Set <i>BOW1</i> ≤ 4096.
	0x2E	23:0		<i>BOW2 (Default: 0x000000)</i>
			U	Bow value 2 (second bow B ₂ of the acceleration ramp).
				Value representation: Frequency mode: [pulses per sec ³] 24 digits and 0 decimal places: $1 \text{ pps}^3 \leq BOW2 \leq 16 \text{ Mpps}^3$ Direct mode: [Δa per clk cycle] $\text{bow}[\text{av per clk_cycle}] = BOW2 / 2^{53}$ $BOW2 [\text{pps}^3] = BOW2 / 2^{53} \cdot f_{\text{CLK}}^3$! Set <i>BOW2</i> ≤ 4096.
	0x2F	23:0		<i>BOW3 (Default: 0x000000)</i>
			U	Bow value 3 (first bow B ₃ of the deceleration ramp).
				Value representation: Frequency mode: [pulses per sec ³] 24 digits and 0 decimal places: $1 \text{ pps}^3 \leq BOW3 \leq 16 \text{ Mpps}^3$ Direct mode: [Δa per clk cycle] $\text{bow}[\text{av per clk_cycle}] = BOW3 / 2^{53}$ $BOW3 [\text{pps}^3] = BOW3 / 2^{53} \cdot f_{\text{CLK}}^3$! Set <i>BOW3</i> ≤ 4096.
	0x30	23:0		<i>BOW4 (Default: 0x000000)</i>
			U	Bow value 4 (second bow B ₄ of the deceleration ramp).
				Value representation: Frequency mode: [pulses per sec ³] 24 digits and 0 decimal places: $1 \text{ pps}^3 \leq BOW4 \leq 16 \text{ Mpps}^3$ Direct mode: [Δa per clk cycle] $\text{bow}[\text{av per clk_cycle}] = BOW4 / 2^{53}$ $BOW4 [\text{pps}^3] = BOW4 / 2^{53} \cdot f_{\text{CLK}}^3$! Set <i>BOW4</i> ≤ 4096.

Table 75: Ramp Generator Registers



17.17. External Clock Frequency Register

External Clock Frequency Register				
R/W	Addr	Bit	Val	Description
RW	0x31	24:0		<i>CLK_FREQ</i> (Default: 0x0F42400)
			U	External clock frequency value f_{CLK} [Hz] with $4.2\text{ MHz} \leq f_{CLK} \leq 30\text{ MHz}$.

Table 76: External Clock Frequency Register

17.18. Target and Compare Registers

Target and Compare Registers				
R/W	Addr	Bit	Val	Description
RW	0x32	31:0		<i>POS_COMP</i> (Default: 0x00000000)
			S	Compare position.
RW	0x33	31:0		<i>VIRT_STOP_LEFT</i> (Default: 0x00000000)
			S	Virtual left stop position.
RW	0x34	31:0		<i>VIRT_STOP_RIGHT</i> (Default: 0x00000000)
			S	Virtual right stop position.
RW	0x35	31:0		<i>X_HOME</i> (Default: 0x00000000)
			S	Actual home position.
R	0x36	31:0		<i>X_LATCH</i> (Default: 0x00000000) <i>(if circular_cnt_as_xlatch = 0)</i>
			S	Storage position for certain triggers.
				<i>REV_CNT</i> (Default: 0x00000000) <i>(if circular_cnt_as_xlatch = 1)</i>
			S	Number of revolutions during circular motion.
W		30:0		<i>X_RANGE</i> (Default: 0x00000000)
			U	Limitation for <i>X_ACTUAL</i> during circular motion: $-X_RANGE \leq X_ACTUAL \leq X_RANGE - 1$
RW	0x37	31:0		<i>X_TARGET</i> (Default: 0x00000000)
			U	Target motor position in positioning mode. ! Set all other motion profile parameters before!

Table 77: Target and Compare Registers



17.19. Pipeline Registers

Pipeline Register				
R/W	Addr	Bit	Val	Description
RW	0x38	31:0	S	X_PIPE0 (Default: 0x00000000): 1 st pipeline register.
	0x39	31:0	S	X_PIPE1 (Default: 0x00000000): 2 nd pipeline register.
	0x3A	31:0	S	X_PIPE2 (Default: 0x00000000): 3 rd pipeline register.
	0x3B	31:0	S	X_PIPE3 (Default: 0x00000000): 4 th pipeline register.
	0x3C	31:0	S	X_PIPE4 (Default: 0x00000000): 5 th pipeline register.
	0x3D	31:0	S	X_PIPE5 (Default: 0x00000000): 6 th pipeline register.
	0x3E	31:0	S	X_PIPE6 (Default: 0x00000000): 7 th pipeline register.
	0x3F	31:0	S	X_PIPE7 (Default: 0x00000000): 8 th pipeline register.

Table 78: External Clock Frequency Register

17.20. Shadow Register

Shadow Register				
R/W	Addr	Bit	Val	Description
RW	0x40	31:0	S	SH_REG0 (Default: 0x00000000) : 1 st shadow register.
	0x41	31:0	U	SH_REG1 (Default: 0x00000000) : 2 nd shadow register.
	0x42	31:0	U	SH_REG2 (Default: 0x00000000) : 3 rd shadow register.
	0x43	31:0	U	SH_REG3 (Default: 0x00000000) : 4 th shadow register.
	0x44	31:0	U	SH_REG4 (Default: 0x00000000) : 5 th shadow register.
	0x45	31:0	U	SH_REG5 (Default: 0x00000000) : 6 th shadow register.
	0x46	31:0	U	SH_REG6 (Default: 0x00000000) : 7 th shadow register.
	0x47	31:0	S/U	SH_REG7 (Default: 0x00000000) : 8 th shadow register.
	0x48	31:0	U	SH_REG8 (Default: 0x00000000) : 9 th shadow register.
	0x49	31:0	U	SH_REG9 (Default: 0x00000000) : 10 th shadow register.
	0x4A	31:0	U	SH_REG10 (Default: 0x00000000) : 11 th shadow register.
	0x4B	31:0	U	SH_REG11 (Default: 0x00000000) : 12 th shadow register.
	0x4C	31:0	U	SH_REG12 (Default: 0x00000000) : 13 th shadow register.
	0x4D	31:0	U	SH_REG13 (Default: 0x00000000) : 14 th shadow register.

Table 79: External Clock Frequency Register

